AI.410

Robot Vision

COURSE OVERVIEW

Vision integrated with robot action: visual servoing, hand-eye coordination, and vision-based manipulation.



ILLUSTRATIVE HERO IMAGERY

ROBOT VISION

1

SESSION

60 MINUTES

ORIENTATION: VISION IN ACTION

KEY TOPICS

- Robot vision vs. machine vision
- Hand–eye coordination
- Vision–action loop
- Applications

TEACHING FOCUS

Let vision guide robot motion.

KEY TAKEAWAY Vision directs the hand.

2

SESSION

60 MINUTES

VISUAL SERVOING

KEY TOPICS

- Image-based vs. position-based
- Visual servo control
- Camera calibration
- Eye-in-hand vs. eye-to-hand

TEACHING FOCUS

Close the loop through the camera.

KEY TAKEAWAY Visual servoing moves by what it sees.

ROBOT VISION

3

SESSION

60 MINUTES

VISION-BASED NAVIGATION

KEY TOPICS

- Visual odometry
- SLAM with cameras
- Monocular and stereo
- Loop closure

TEACHING FOCUS

Navigate using vision alone.

▲ **KEY TAKEAWAY** Cameras are lightweight eyes.

4

SESSION

60 MINUTES

VISION-BASED MANIPULATION

KEY TOPICS

- Guided grasping
- Object tracking for manipulation
- Vision-tactile fusion
- Assembly by vision

TEACHING FOCUS

Manipulate with visual feedback.

▲ **KEY TAKEAWAY** Vision guides manipulation precisely.

ROBOT VISION

5

SESSION

60 MINUTES

ROBOT VISION SYSTEMS

KEY TOPICS

- Pipeline design
- Real-time constraints
- Calibration and maintenance
- Evaluation metrics

TEACHING FOCUS

Integrate vision on robots.

KEY TAKEAWAY Robot vision must be fast and robust.

ROBOT VISION

6

SESSION

60 MINUTES

ROBOT VISION & REVIEW

KEY TOPICS

- Vision–robot demo
- Course recap and assessment
- Toward robot design (AI.411)
- Vision task

TEACHING FOCUS

Combine sight and motion on a robot.

KEY TAKEAWAY Sight plus action equals robot vision.